BullSO 2026 Invitational

Circuit Lab B

Bull Science Olympiad at the University of South Florida

Saturday, January 24th, 2026

Team Number: _____

Team Name:_____

Participant Names:_____

Total Score:_____/100 **Rank:_____**

PART I: Multiple Choice (41 pts)

1. The electrostatic force between two charges is directly proportional to the product of their charges and inversely proportional to the square of the distance between them. Who was this law named after? (1 pt)

   A. Ampere B. Coulomb C. Volta D. Faraday

2. Who invented the electric battery? (1 pt)

   A. Ohm B. Coulomb C. Volta D. Kirchoff

3. Who coined the term “electrode”? (1 pt)

   A. Coulomb B. Volta C. Faraday D. Tesla

4. The SI unit for magnetic flux density is named after who? (1 pt)

   A. Ampere B. Volta C. Faraday D. Tesla

5. The law stating the relationship between voltage, current, and resistance was named after who? (1 pt)

   A. Ampere B. Ohm C. Volta D. Tesla

6. Who was named “the Newton of electricity” in James Clerk Maxwell’s *A Treatise on Electricity and Magnetism*? (1 pt)

   A. Ampere B. Kirchoff C. Volta D. Tesla

7. Which of the following are NOT passive components? (Select ALL correct answers) (2 pts)

   A. Transistor B. Inductor C. Capacitor D. Diode E. Resistor

8. A magnetic field in the -x direction causes an electron to experience a magnetic force in the +y direction. The electron is moving in which direction? (1 pt)

   A. +x B. -y C. Out of page D. Into the page

9. Suppose a wire of length L has resistance R. If the same wire's length is tripled and its cross sectional area is halved, what is the new resistance in terms of R? (1 pt)

A. 0.5R B. 1.5R C. 3R D. 6R

10. Which of the following is NOT a way to increase a capacitor's capacitance? (1 pt)

A. Increase distance between the plates
B. Use a dielectric with a lower permittivity
C. Use a dielectric with a higher permittivity
D. Increase surface area of the plates

11. A point charge Q_1 = +6 μC is separated from a point charge Q_2 = -7 μC by 0.5 meters. What is the electrostatic force between both charges and is the force attractive or repulsive? Use k = $9 \cdot 10^9 \frac{N \cdot m^2}{C^2}$ for Coulomb's Constant. (2 pts)

A. -1.512, Attractive
B. -1.512, Repulsive
C. -0.01512, Attractive
D. -0.01512, Repulsive

12. Resistors in parallel have the same __ and resistors in series have the same __? (2 pts)

A. Voltage; Voltage
B. Voltage; Current
C. Current; Voltage
D. Current; Current

13. Which device is used to convert DC to AC? (1 pt)

A. Capacitor B. Transformer C. Rectifier D. Inverter

14. In a conductor, AC current density inside a conductor is greatest near the surface of the conductor at high frequencies. What is this known as? (1 pt)

A. Edge Effect
B. Hall Effect
C. Skin Effect
D. Proximity Effect

15. What do electric field lines best describe? (1 pt)

A. The direction of force applied on a positive charge due to electrostatic forces
B. The direction of force applied on a negative charge due to electrostatic forces
C. The path that negative charges take when placed on the electric field lines

D. The path that positive charges take when placed on the electric field lines

16. What is the relationship between electric field lines and equipotential lines? (1 pt)

A. The lines are parallel to each other
B. The lines are perpendicular to each other
C. The lines can intersect at any angle
D. The lines have no relevant relationship with each other

17. A charge drifts away from a point charge and positive work is done on the charge to place it back to a point which lies on the same equipotential line as the charge's initial placement. Which of the following is true? (Select ALL correct answers) (2 pts)

A. The charge has undergone a net change in electric potential energy
B. The charges are alike in sign
C. The final position of the charge must be identical to the initial placement
D. The work done on the charge is stored in the electric potential energy of the system

18. Which of the following cannot exist? (1 pt)

A. A room temperature superconductor
B. A magnetic monopole
C. A magnetic quadrupole
D. Forever circulating current in a superconductor

19. Which of the following can transformers not change? (1 pt)

A. Voltage
B. Current
C. Resistance
D. Maximum power

20. What is the ratio of Secondary winding turns to Primary winding turns proportional to? (1 pt)

A. The ratio of power output to input
B. The ratio of current output to input
C. The ratio of inductance output to input
D. The ratio of voltage output to input

21. Are batteries able to output AC current? (1 pt)

A. Yes B. No

22. What DC voltage is needed to provide the same power as a 100V sinusoidal AC source? (1 pt)

A. $100/(\sqrt{2})$ B. $100\sqrt{2}$ C. 200 D. 50

23. If an electromagnet was made of superconducting material and injected with an initial current, would it be able to sustain a magnetic field forever? (1 pt)

A. No, because generation of a magnetic field requires an oscillating current
B. Yes, because magnetic fields are self sustaining, for example, natural magnets
C. No, because superconductors decay current over time
D. Yes, because the superconductors resist decay of current forever

24. How does a three-way switch setup operate? (1 pt)

A. The switch has three positions, closed, open, and short
B. The circuit contains three switches, each able to change the state of a lightbulb
C. The circuit contains two switches, each able to change the state of a lightbulb
D. The switch is able to switch between three different wires, creating 3 different possible paths from one switch

25. If a positive charge is placed with no initial velocity in a uniform magnetic field, how will it behave? (1 pt)

A. The charge will be pushed in the direction of the magnetic field
B. The charge will be pushed perpendicular to the magnetic field
C. The charge will remain at rest
D. There is not enough information

26. If a negative charge is placed in a uniform magnetic field and given an initial velocity, how will it behave? (1 pt)

A. It will remain traveling with its initial velocity
B. It will turn 90 degrees due to the force applied from the magnetic field
C. It will move in circles
D. There is not enough information

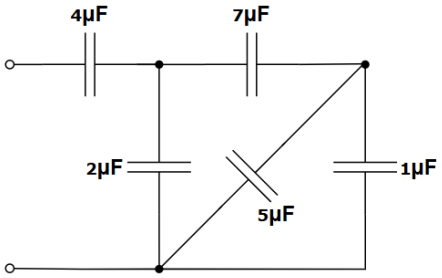
27. If a 1 Coulomb positive charge moves perpendicular to a 1 tesla magnetic field at 1 meter per second, what work will the magnetic field do on the charge? The mass of the charge is 1 kg. (1 pt)

A. $\sqrt{2}$ J
B. $\sqrt{2}$ electron volts
C. 1 electron volts
D. 0 electron volts

28. What does GFCI stand for (GFCI outlet)? (1 pt)
A. Galton Frances Combined Interruptor
B. Ground Fault Circuit Interruptor
C. General Friction Current Injector
D. Granular Form Clustered Initiator

29. What is a major difference between Fuses and Breakers? (Select ALL correct answers) (2 pts)
A. Only one is suited for overcurrent protection
B. Only one is suited for fault interruption
C. Only one is able to be reset without replacement
D. Only one is suitable for alternating current systems

30. Consider the following circuit.



What is the equivalent capacitance? (2 pts)

A. 2μF B. 0.5μF C. 4μF D. 8μF

31. Suppose a parallel plate capacitor is connected across a 12 V battery and the magnitude of charge stored on one plate is 804 Coulombs. What is the capacitor's capacitance? (1 pt)

A. 12 F B. 67F C. 134 F D. 9648 F

32. Which of the following best indicates where the energy of a capacitor is stored and in what form? (2 pt)

A. The electric field, potential energy.
B. The magnetic field, potential energy.
C. The charge in the plates, potential energy.
D. The current, kinetic energy

33. Which of the following practices is least practical at preventing shocking of the user of appliances with conductive metal cases containing circuitry? (1 pt)

A. Using three pronged plugs to ground the case.
B. Designing the appliance to minimize contact between the hot wire and the case.
C. Using polarized plugs to control which wire is hot.
D. Incorporating an insulator in extra space in the case.

34. Which of the following is not a possible effect of experiencing current flow through the body? (1 pt)

A. Weakening of muscles in the brain, increasing susceptibility to successive shocking.
B. Internal tissue burning due to the electrical resistance of the body.
C. Fibrillation of the heart due to shock to the diaphragm muscle.
D. Interference from the current with nerve cell communication, causing clenching of muscles.

PART II: Short Answer (30 pts)

1. State the name of the following circuit components. (1 pt each)

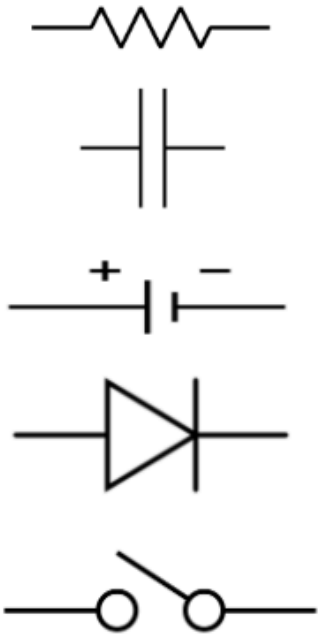
a. _____

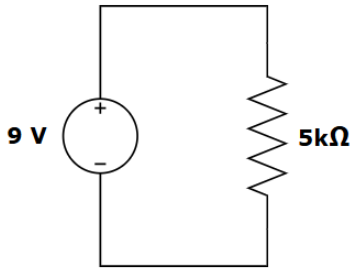
b. _____

c. _____

d. _____

e. _____



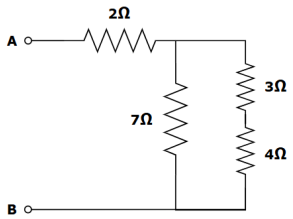


2. Consider the circuit shown above.
   a. Calculate the current through the 5kΩ resistor and include units. (1 pt)

   b. Calculate the power delivered by the 9 V voltage source and include units. (1 pt)

   c. To produce a current of 3 mA, you should replace the 5kΩ resistor with a resistor of how many Ohms? (1 pt)

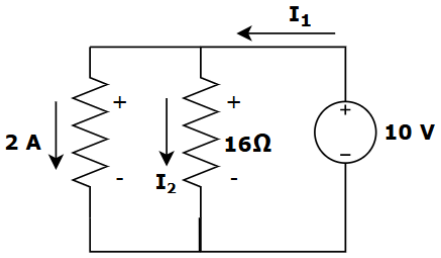
Use the circuit shown below for questions 3 and 4.



3. Calculate the equivalent resistance between points A and B and include units. (2 pts)

4. If all 4 resistors in the circuit were replaced with 4 resistors of the same resistance, what resistance would result in an equivalent resistance of 45Ω? (2 pt)

5. Consider the following circuit.

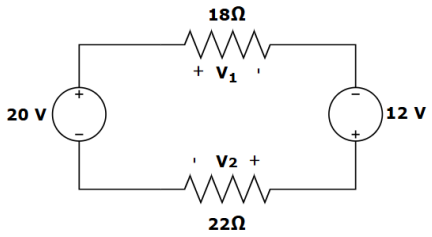


a. What law should be used to solve for I_1? (1 pt)

b. What law should be used to solve for I_2? (1 pt)

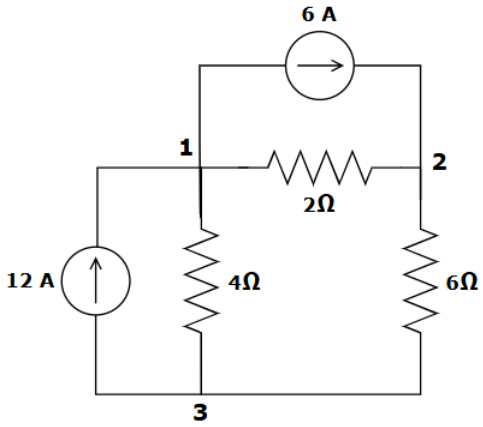
c. Solve for I_1 and I_2 and include units. (4 pts)

6. Consider the following circuit.



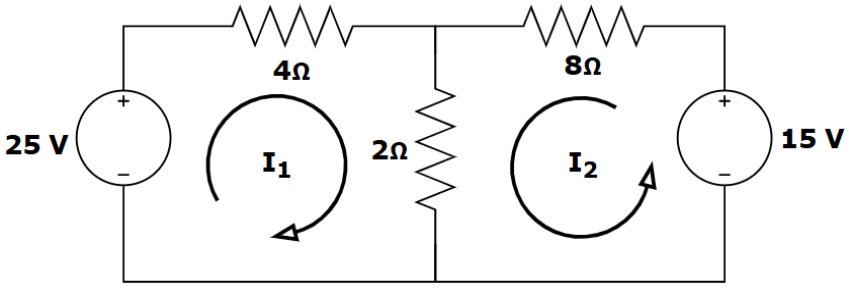
Determine the voltages V_1 and V_2. Clearly show all your work and include units. (4 pts)

7. Consider the following circuit.



Using nodal analysis, determine the node voltages V_1 and V_2 at nodes 1 and 2 given that node 3 is the reference node (ground). Clearly show all work and include units. (4 pts)

8. Consider the following circuit.



Using mesh analysis, determine the mesh currents I_1 and I_2. Clearly show all your work and include units. (4 pts)

Division B Circuit Lab - Lab Portion - 29pts

You are given a breadboard, 4 resistors, and jumper wires.

The resistor values are:

- R_1 = 100 Ω
- R_2 = 200 Ω
- R_3 = 50 Ω
- R_4 = 50 Ω

Task

Using all four resistors exactly once, construct a circuit with an equivalent resistance of 100 Ω between the two output terminals.

At least two resistors must be connected in parallel and at least one resistor must be in series with another combination.

What to submit

1. A clearly labeled sketch of your circuit. - 8pts
2. Step-by-step calculations showing intermediate equivalent resistances. - 8pts
3. A short explanation of why your configuration produces **100 Ω**.- 4pts
4. _____ Event Supervisor signs off for 100 Ohm configuration - 9pts

BullSO, University of South Florida - Repaired practice edition

Division B | Circuit Lab | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

Name: _____ Date: _____

repair-case. (4 points) Authored written substitute for unavailable source material: Why test all input combinations?

Unit: Digital logic and combinational circuits.
Worked-case data: For A=1, B=1, find half-adder sum and carry.

Answer: _____
_____
_____